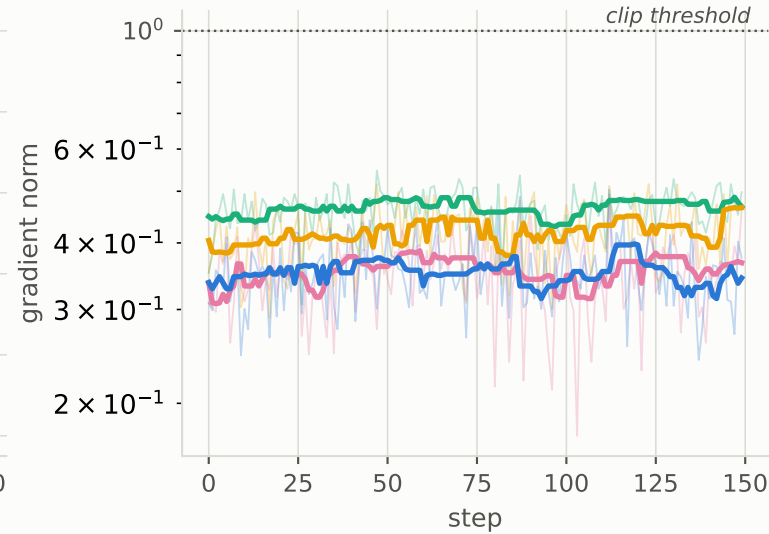
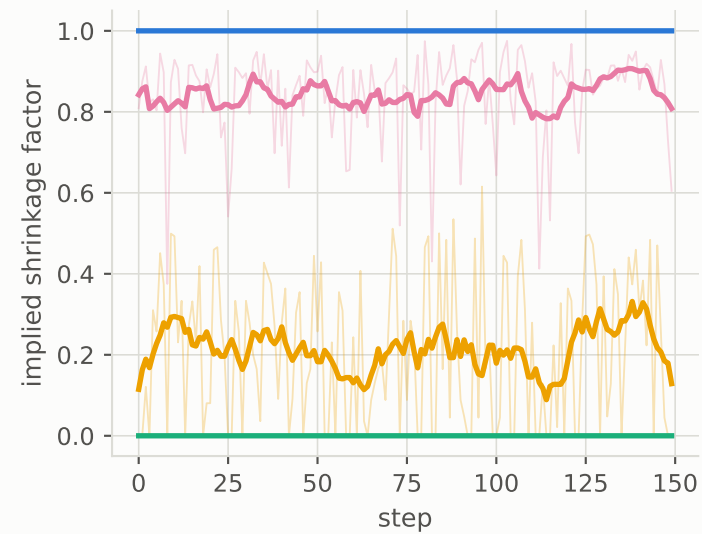
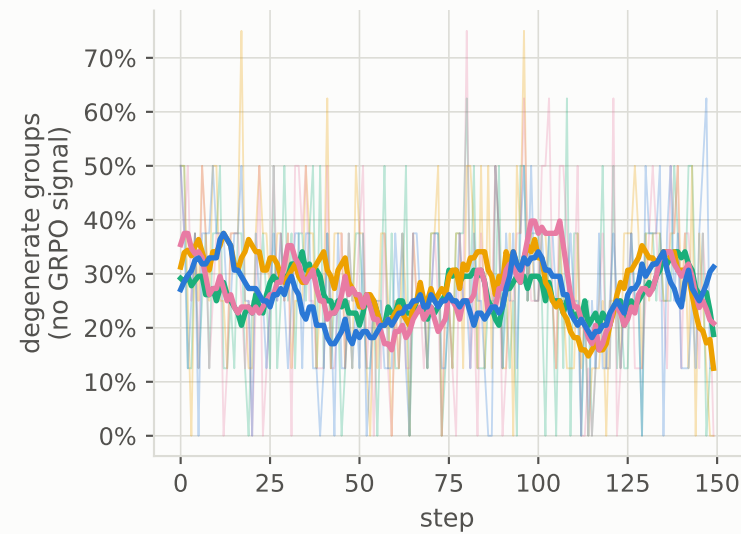
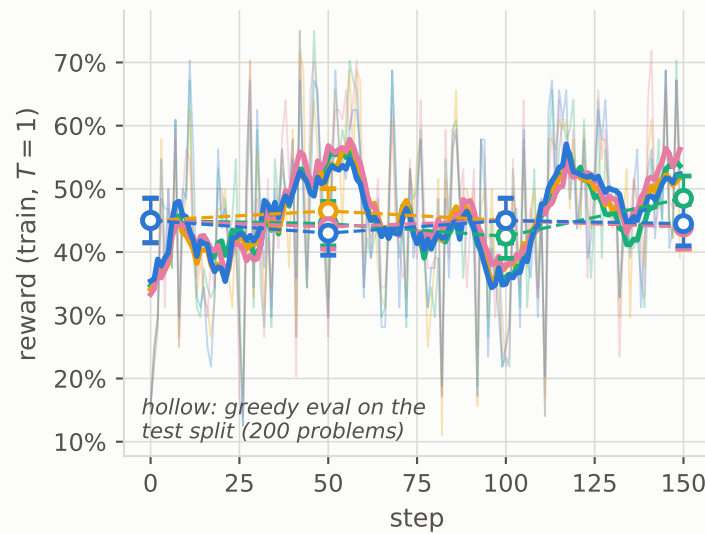


GRPO on GSM8K — Qwen2.5-0.5B-Instruct, K=8 x N=8, lr=1e-06, seed 0, faint = per step, bold = rolling window of 10



— global mean — JS, $V = 1/N$ (GRPO.md) — JS, V estimated — GRPO (group mean)